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SUPERSEDING
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MILITARY SPECIFICATION

# ABRASIVE, DENTAL, POWDER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers an abrasive powder suitable for cleaning and polishing natural teeth.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-566 | - Boxes, Folding, Paperboard. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-B-676 | - Boxes, Setup. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 576 - Chemical Analysis of Zircon Refractories.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington DC 20204.)

3. REQUIREMENTS

3.1 Material. The material shall be essentially 90 percent zirconium silicate crystals and 10 percent tin oxide, flavored with water soluble spearmint and saccharine (see 3.3.1). Sweetening agents used in the formulation of the paste shall be free from cyclamates. The zirconium silicate breakdown shall comply with and be reported on the following basis (see 6.2):

| | |
|---|---|
| $ZrO_2$ | 65.0 percent (minimum) |
| $SiO_2$ | 34.0 percent (maximum) |
| $TiO_2$ | 0.65 percent (maximum) |
| $Fe_2O_3$ | 0.20 percent (maximum) |
| Free Silica | 0.50 percent (maximum) |



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Percentage of zirconium silicate and tin oxide shall be ascertained from the manufacturer's production control records. Materials shall be of the highest quality throughout and suitable in all respects for the intended purpose.

3.2 Description. The crystals shall be prepared by a hammer mill process, which produces a roughened, irregular surface configuration. Each unit box shall contain a suitable translucent plastic pipet with a line marked and calibrated to deliver 0.7 ± 0.05 milliliter (ml) of water, and a double-ended measuring device to deliver 2.0 ± 0.10 grams of compatible lava pumice and 0.35 ± 0.05 gram of stannous fluoride.

3.3 Physical properties.

3.3.1 Flavoring. The flavoring agents shall impart a pleasant taste to the abrasive powder. The flavoring agents shall have no significant adverse effect on the cleaning or polishing properties of the preparation.

3.3.2 Particle size. The particle size distribution of the zirconium silicate shall be as follows, when tested as specified in 4.4.2:

| Particle size | Percentage range |
|---|---|
| Less than 10 microns | 0 to 10 percent |
| 10 to 20 microns | 10 to 30 percent |
| 20 to 30 microns | 15 to 35 percent |
| 30 to 44 microns | 15 to 35 percent |
| 44 to 74 microns | 15 to 35 percent |
| 74 to 90 microns | 0 to 10 percent |
| Over 90 microns | none |

3.3.3 Polishing ratio. The polishing ratio shall be 14 to 22, when tested immediately after mixing.

3.4 Irritation. Prior to delivery, the manufacturer shall furnish a statement directly to the Technical Services Branch, (Code ATTS-2), Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, stating the abrasive powder does not normally irritate oral tissues and does not contain poisonous ingredients in sufficient concentration to be harmful to human beings when used as directed.

3.5 Weight. Each cup shall contain a minimum of 4.5 grams of abrasive powder. Each unit box (containing 100 cups) shall contain a minimum of 500 grams of powder.



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3.6 Performance. The abrasive powder shall adhere to the prophylactic brush and cup, and shall be capable of cleaning difficult stains, leaving a smooth polished surface on the dental hard tissues.

3.7 Instructions for use. Adequate and accurate instructions for the preparation and use of the abrasive powder as a prophylaxis cleaner and polishing agent shall accompany each unit box. In addition, two copies of the instructions shall be forwarded to the contracting officer for information purposes.

3.8 Unit of issue. Each unit box shall consist of 100 cups of abrasive powder containing a minimum of 500 grams of powder, one pipet and measuring device (see 3.2), and instructions for preparation and use (see 3.7).

3.9 Delivery. Not more than 6 months shall have elapsed from the date of packaging of the abrasive powder to the date of delivery to the Government.

3.10 Labeling. Labeling of the cup shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act.

3.11 Workmanship. Workmanship shall be first class throughout. The abrasive powder shall be free from defects which detract from its appearance or impair its serviceability.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.



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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials or packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Manufacturer's records. The percentage of zirconium silicate and tin oxide shall be ascertained from the manufacturer's production control records.

4.2 Sampling

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level S-1, acceptable quality level (AQL) 2.5 (percent defective) for major defects and 4.0 (percent defective) for minor defects. Unit of product for examination shall be one immediate container of abrasive powder.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with table I. Unit of product for test purposes shall be one immediate container of abrasive powder.

TABLE I. Sampling for tests.

| Characteristics | Requirements | Test procedure | Unit of product to be tested per lot or batch no. | Number of determinations per unit of product | Total number of determinations |
|---|---|---|---|---|---|
| Flavoring | 3.3.1 | 4.4.1 | 1 | 1 | 1 |
| Particle size | 3.3.2 | 4.4.2 | 3 | 1 | 3 |
| Polishing ratio | 3.3.3 | 4.4.3 | 2 | 1 | 2 |
| Weight | 3.5 | SMI 1/ | 5 | 1 | 5 |
| Performance test | 3.6 | 4.4.5 | 1 | 1 | 1 |

1/ Standard Measuring Instruments



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4.2.3 Composition of zirconium silicate (raw material). A sufficient amount of zirconium silicate shall be taken from each lot or batch number (before it is combined with the other raw materials to make the abrasive powder) to determine composition (see 3.1 and 4.4.4).

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The abrasive powder shall be examined for defects including, but not limited to, those listed in table II.

TABLE II. Classification of defects.

| Categories | Defects |
|---|---|
| Major | |
| 101 | Components missing in unit box. |
| 102 | Powder is not free flowing. |
| 103 | Powder contains dirt or other foreign material. |
| 104 | Instructions not adequate or accurate. |
| 105 | Pipet not as specified. |
| 106 | Measuring device not as specified. |
| Minor | |
| 201 | Less than 100 cups in each unit of issue. |
| 202 | Delivery schedule not as specified. |
| 203 | Labeling of cup not as specified. |

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Flavoring. The pleasantness or unpleasantness of the flavor shall be determined by tasting the material.

4.4.2 Particle size. Particle size distribution shall be determined by flotation or aqueous sedimentation procedures (see 6.3). As an alternate, particle size of 44 microns or greater shall be determined by screening.

4.4.3 Polishing ratio. The polishing ratio shall be determined by comparison of abrasive powder to a standard calcium carbonate (see 6.3).



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4.4.4 Zirconium silicate composition. Chemical analysis of the zirconium silicate shall be determined by American Society for Testing and Materials (ASTM) method C 576 with modifications as required or other recognized means of analysis.

4.4.5 Performance test. Dental abrasive powder shall be used in accordance with the instructions (see 3.7) and shall be given a practical performance test to determine compliance with the specified performance requirements cited in 3.6.

4.5 Packaging inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One box containing 100 immediate containers (cups), one pipet, measuring device, and instructions for use, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Immediate container. Immediate container shall be a flat shouldered plastic container having an easily removed cover. Each container shall be capable of containing 5 grams of powder.

5.1.2.2 Unit package. Twenty-five immediate containers (cups) attached shoulder to shoulder shall form a tray from which individual containers (cups) may be snapped off for use. Four trays shall be packaged in a unit package constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic, or PPP-B-676. Box design shall include top, bottom, and tier pads fabricated of the same material as the box. Closure shall be as specified in the appendix of the box specification. In addition, each unit package shall contain one pipet, measuring device, and a brochure containing complete instructions for preparation and use with other agents used in preventive dentistry.

5.1.3 Level C. One hundred containers of abrasive powder containing a minimum of 500 grams of powder, one pipet, measuring device, and instructions for use shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.



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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Forty-eight unit packages shall be packaged in a shipping container designed for a type 2 load and conforming to PPP-B-636, classs weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Forty-eight unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged powder shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking

5.3.1 Immediate container. Each immediate container shall be marked in accordance with the contractor's commercial practice.

5.3.2 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number, and name and address of manufacturer shall be shown. In additon, marking shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act.

5.3.3 Shipping container. Shipping containers shall be marked in accordance with MIL-STD-129. The lot (control) number shall be shown.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).



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6.2 Symbols. The following chemical symbols were used herein to indicate elements (see 3.1):

Zr - Zirconium
Si - Silicon
Ti - Titanium
Fe - Iron
O - Oxygen

6.3 Literature reference. Studies concerning the Polishing Properties of Zirconium Silicate on Enamel by George Stooky, James R. Hudson, and Joseph C. Muhler, The Journal of Periodontology, Vol. 37, Pages 200-207, May-June, 1966.

6.4 This specification covers the following item appearing in the Federal Supply Catalog: NSN 6520-00-926-8873 ABRASIVE, DENTAL, Powder, Zirconium Silicate, 5 Gram, 100s.

6.5 Limitations. This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD-MB

Preparing activity:
DLA-DM

Project No. 6520-1485



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-36826A Abrasive, Dental, Powder

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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2. REMARKS

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